MEA(R)N Full Stack Programming

byteXL

Overview of Web Development

Web development support to create, build, and maintain websites. It also covers database management, web development, web publishing, and web design. It's developing an online application, such as a website.

“Web Development" is a combination of two words:

- Web: This term describes webpages, websites, and anything that operates online.
- Development: This is the process of creating an application from the ground up.

Classification of web Development

01	FrontEnd Development	• It focuses on the **visual and interactive aspects of a website**. • Also called **client-side development.** • To build UI, Use **HTML, CSS, and JavaScript** • It is used to create an **engaging and responsive user experience.**
02	Backend Development	• It involves creating the **server** and **database** components of a website or app. • Back-end developers manage **application logic, data storage,** and **server communication**. • Also called **Server-side development.**

Road Map of Full Stack Developer

FrontEnd

Basic
- HTML
- CSS
- JavaScript

Framework
- React
- VUE
- Angular

Styles
- Material UI
- Bootstrap

BackEnd

Technology
- PHP
- NodeJS
- Ruby
- Java
- Python
- Redis
- NET core

Database

RDBMS
- Oracle, MSSQL, MySQL

NoSQL
- MongoDB, Casandra, CouchDB

Graph
- Neo4j, Bootstrap

Message Queue
- Kafka, SQS , RabbitMQ

DevOps

Infrastructure
- AWS, Azure, GCP

Automation
- Jenkins, Chef, Ansible

Virtualization
- Docker, Kubernetes, Vagrant, VMware

Agenda

- Fundamentals of Web Application
- Single Page Application
- Full Stack Development

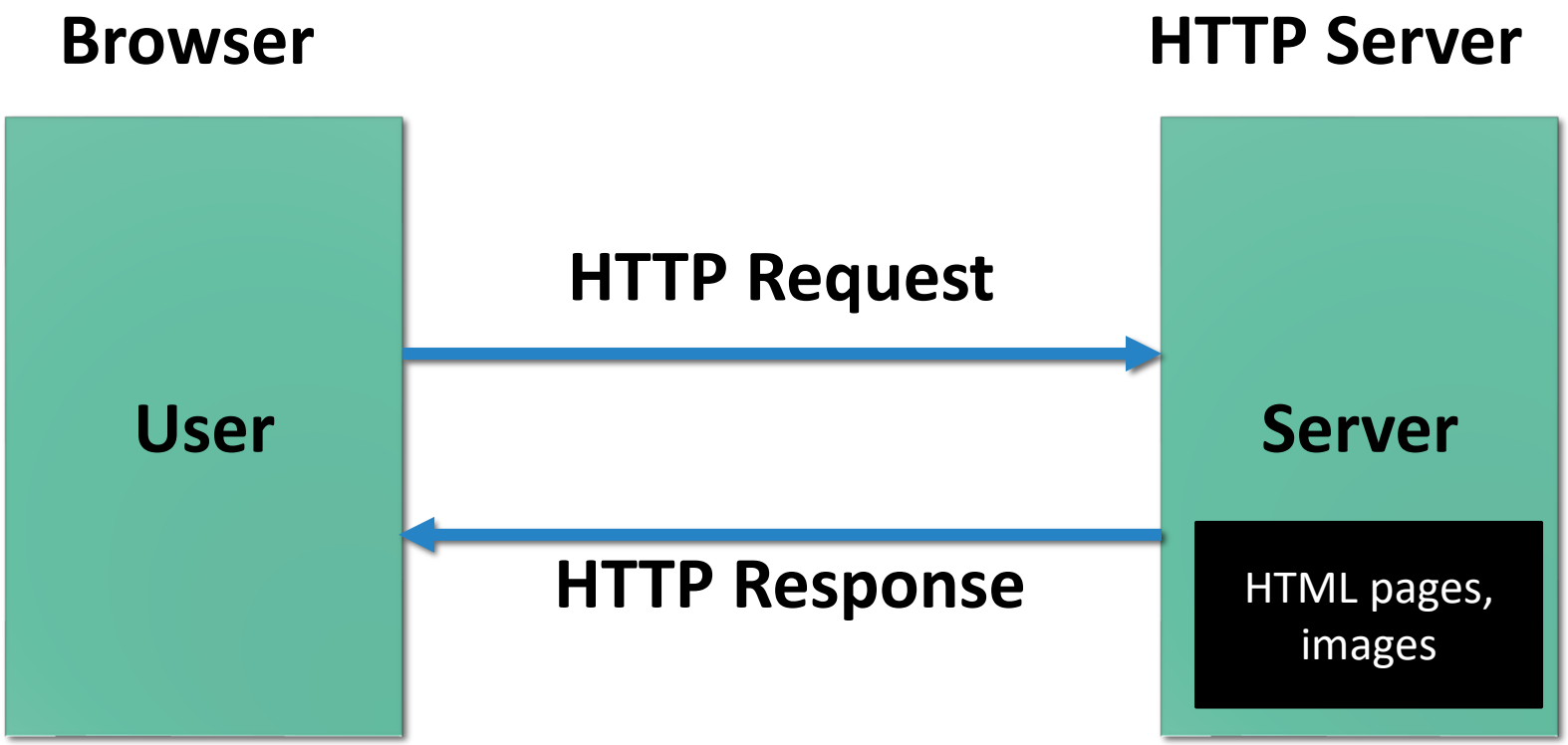
Fundamentals of Web Application

- Introduction to Web Application
- Types of Web Application
- Introduction to Web Application Architecture
- How does Web Application Architecture work ?
- Components of Web Application Architecture
- Web Application Architecture Models
- Types of Web Application Architecture

Introduction to Web Application

- Web Application (Web App) is an Application Software that is stored on a remote server and delivered over the Intranet or Internet through a Browser application.
- Web apps are increasingly popular due to their cross-platform compatibility.
- It can be run on any device with an internet connection and a web browser, including mobile devices.
- Web application retrieve and store information by using server-side scripts, while client-side scripts (in JavaScript and HTML5) present the relevant information on the user interface.

Request / Response Diagram



How do Web Application work ?

1. A user sends a request to a web server via the internet, using either a web browser or the user interface of an app.
2. The web server forwards the request to the appropriate web application server.
3. The web application server then performs the requested task (for example, processing new data) requested of it, and then generates the appropriate results.
4. Results are then sent, with the requested information, from the web application server to the webserver.
5. The web server then responds back to the client and the requested information is displayed to the user.

Types of Web Application

Web applications can be broadly classified into two types:

1. Static Web Application
2. Dynamic Web Application

These classifications are based on how the application handles and delivers content to the user.

Types of Web Application

Static Web Application

- Static web applications are the simplest type of web application.
- They deliver the same pre-stored content to every user.
- The content of a static web application doesn’t change in response to user input. Instead, any changes to the content must be made directly in the HTML code by a developer.
- Static web apps are typically built using ***HTML***, ***CSS***, and sometimes ***JavaScript***.

Types of Web Application

Dynamic Web Application

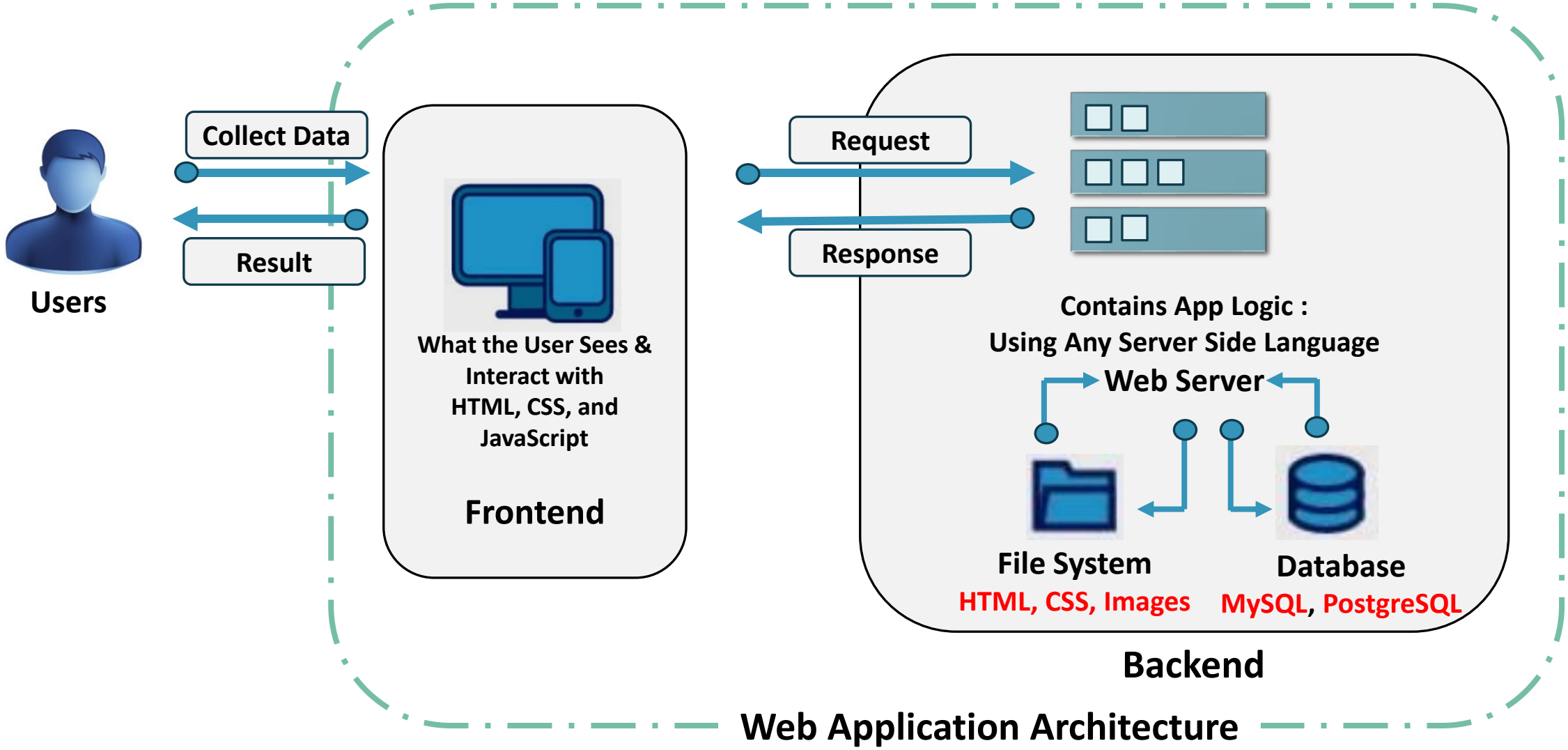
- It delivered the different content based on user interaction or other factors.
- They are built using **server-side *languages*** like PHP, Python, Ruby, or Java, along with a database to store and retrieve data.
- Dynamic web applications are more complex than static ones but more powerful.
- they are best suited for websites that require user interaction or real-time updates, such as social media platforms, e-commerce sites, or online forums.

What is Web Application Architecture?

- Web application architecture is a layout of simultaneous interaction between various web application components:

  User Interfaces, Middleware Component, Databases, and servers.
- It provides the flow and communication of the entire application and handles the major aspects and components of the product.
- It also defining the inter-relations between a swarm of containerized backend servers, load balancers, API gateways, and user-facing single-page frontends.
- It can also be described as the connection between the client and server that defines the connection, along with the server that handles the communication between the client and server.

How Does the Web Application Architecture Work?



Components of Web Application Architecture

- Components are the core part of web application architecture.
- It determines the functionalities of your application and the quality of the user experience offered to the users.
- The web application architecture are categorized into two types:
  1. **UI/UX components (client-side)**
  2. **Structural components (server-side)**

Components of Web Application Architecture

1. **UI/UX components (client-side)**
   - It includes the interface elements, input controls, navigational components, dashboards, statics and analytical elements, admin panel, activity tracking, tools, design layouts, and many more.

2. **Structural components (server-side):**
   - It includes database storage and server control. As the name suggests, it focuses on the structural part of the web app.
   - It mainly comprises of:

     Web (client) browser, Web application server, and Database server

Web Application Architecture Models

The web application architecture models are categorized into two types:

1. **One database, One Web Server**
2. **One database, Multiple Web Servers**
3. **Multiple databases, Multiple Web Servers**

Web Application Architecture Models

1. **One database, One Web Server**
   - It is the most basic and simple model used in web application architecture models.
   - Developers use only a single database and web server to develop the application.
   - It is mainly used in mobile application architecture.
   - **It has one limitation:** if the server is down, then the whole application will be down. Therefore, the model is less preferred for web application.

Web Application Architecture Models

2. One database, Multiple Web Servers

- It is best for those applications where data storage is not done using web servers. Instead, the data is processed by the web server and stored over the database.
- This is a stateless architecture that consists of at least two web servers.
- **High availability.** For example, whenever one server fails due to some reason, the second server would take its place and executes the requests.

Web Application Architecture Models

3. Multiple databases, Multiple Web Servers

- It is the most efficient model that ensures no point of failure, neither for web servers nor databases.
- The models contain two kinds of database models that store identical data and are distributed evenly.

Types of Web Application Architecture

1. Single Page Applications (**SPAs**)
2. Server-Side Rendering (**SSR**)
3. Static Site Generation (**SSG**)
4. Progressive Web Application (**PWA**)

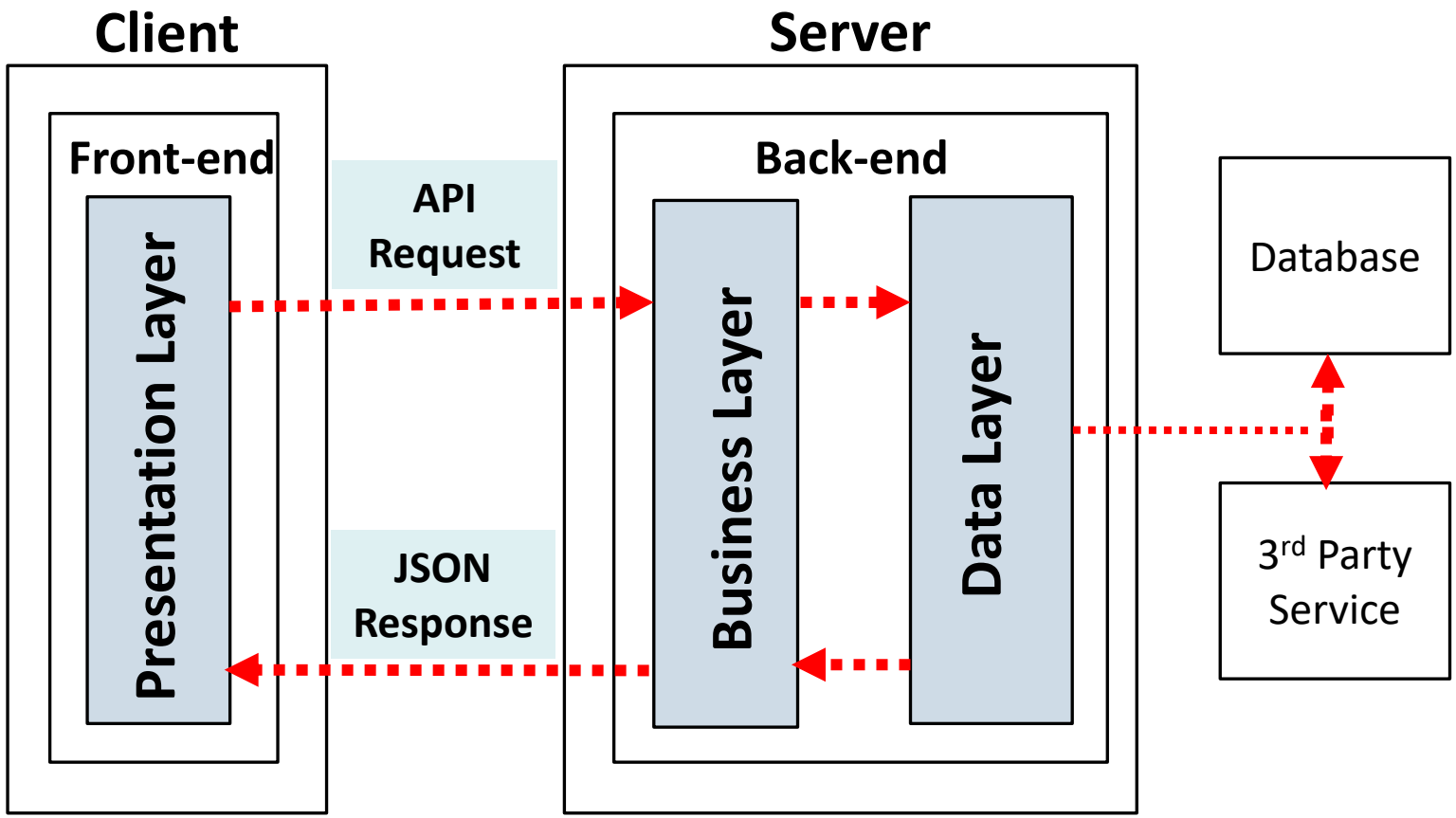
Types of Web Application Architecture

Single Page Applications (SPAs)

- Single Page Application architecture (SPA) redraws any part of the UI without requiring a server roundtrip to retrieve HTML.
- it doesn’t require reloading the page when you need new data.
- **Example** : Facebook, Google Maps, Gmail, GitHub, and Twitter are the best-known examples of single-page apps.

Types of Web Application Architecture

Single Page Applications (SPAs)



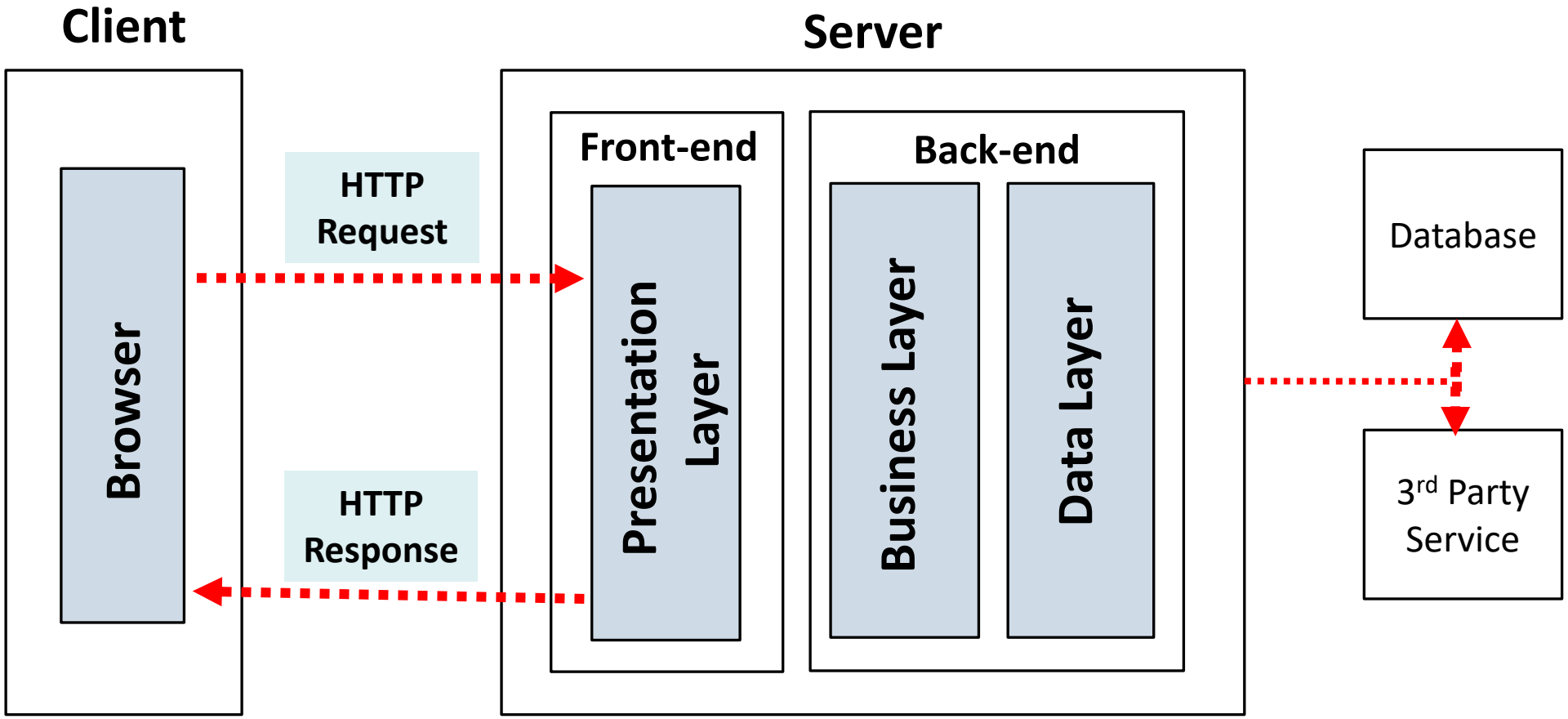
Types of Web Application Architecture

Server-Side Rendering (SSR)

- Server-Side Rendering (SSR) allows an application to convert HTML files on the server into a fully rendered HTML page for the client.
- The web browser submits a request for information from the server that responds instantly by sending a fully rendered page to the client.
- **Example** : Amazon, New York Times, Twitter, LinkedIn, Netflix, and indeed are the best-known examples of Server-Side Rendering.

Types of Web Application Architecture

Server-Side Rendering (SSR)



Types of Web Application Architecture

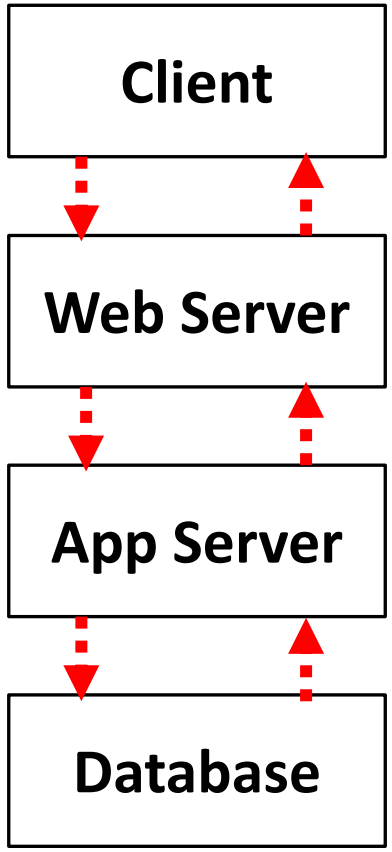
Static Site Generation (SSG)

- Static Site Generation (SSG) requires a generator that automates the coding of individual HTML pages, creating them from templates.
- With SSG, you get a simple static website located on a content delivery network (CDN), or any server, that holds an already generated HTML page to be given to users upon request.
- **Example :** Blogs and Portfolios, Documentation Site, Marketing Pages, and E-Commerce are the best-known examples of Static Site Generation.

Types of Web Application Architecture

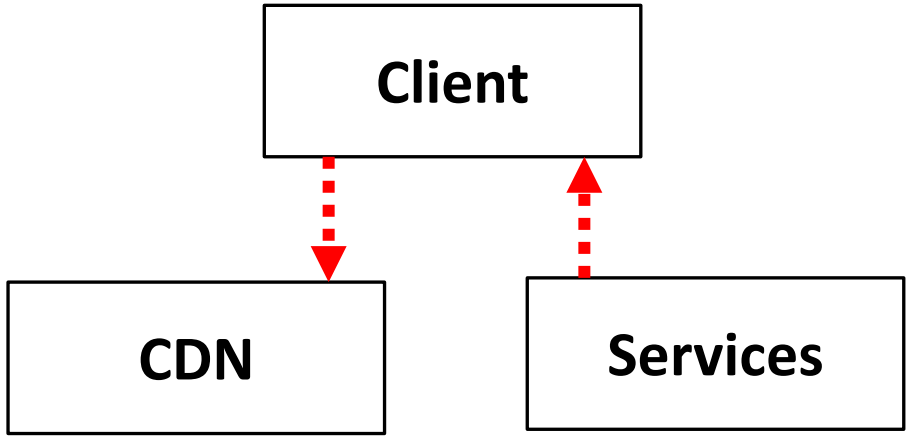
Static Site Generation

Legacy Web



The site is built every time when it is requested by the user

Modern Web



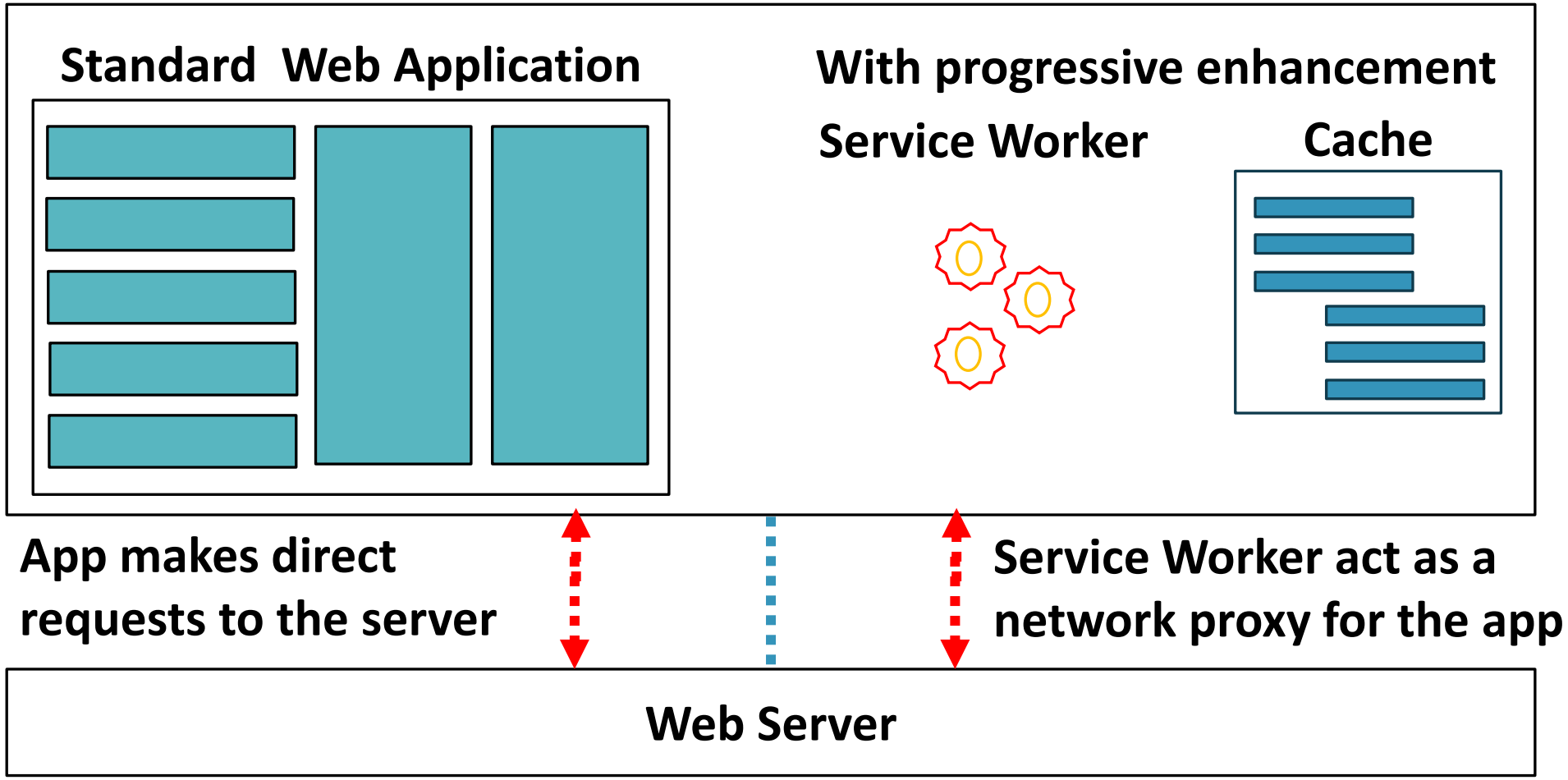
Types of Web Application Architecture

Progressive Web Application

- It act like mobile apps - they can be installed on your phone or desktop and work just like a native app / cross platform app.
- It load quickly and give users a smooth, app-like experience without needing to download from an app store.
- They are built using web technologies such as HTML, CSS, JavaScript, and Web Assembly.
- **Example:** Twitter Lite, Pinterest, and Spotify Web Player

Types of Web Application Architecture

Progressive Web Application



Single Page Application

- Introduction to SPAs
- Advantages and Disadvantages of SPAs
- How do SPAs work ?
- Use cases of SPAs
- Web Frameworks for developing SPAs

Introduction to SPAs

- A Single Page Application (SPA) is a single web page, website, or web application that works within a web browser and loads just a single document.
- It does not need page reloading during its usage, and most of its content remains the same while only some of it needs updating.
- Server used JavaScript APIs when the content needs to be updated.
- It offers a more dynamic web experience to the users.
- It increases the application performance.
- Example : Netflix, Facebook, Trello, Gmail, Google Maps etc.

Advantages of SPAs

Team Collaboration

- Multiple developers is working on same project simultaneously.
- Back-end developers focus on API.
- Front-end developers focus on creating the user interface based on the backend API.

Caching

- A single-page application can cache local data effectively.
- SPA directs only one request to the server and then stocks all the data it obtains, then it can utilize this data and function even while being offline.

Advantages of SPAs

Fast and responsive

As only parts of the pages are loaded dynamically, it improves the website's speed.

Debugging is easier

Debugging single-page applications with chrome is easier since such applications are developed using AngularJS and React developer tools.

Linear user experience

Browsing or navigating through the website is easy.

Disadvantages of SPAs

Browser History Not Stored

- It does not save the users' transition of states within the website.
- A browser saves the previous pages only, not the state transition.

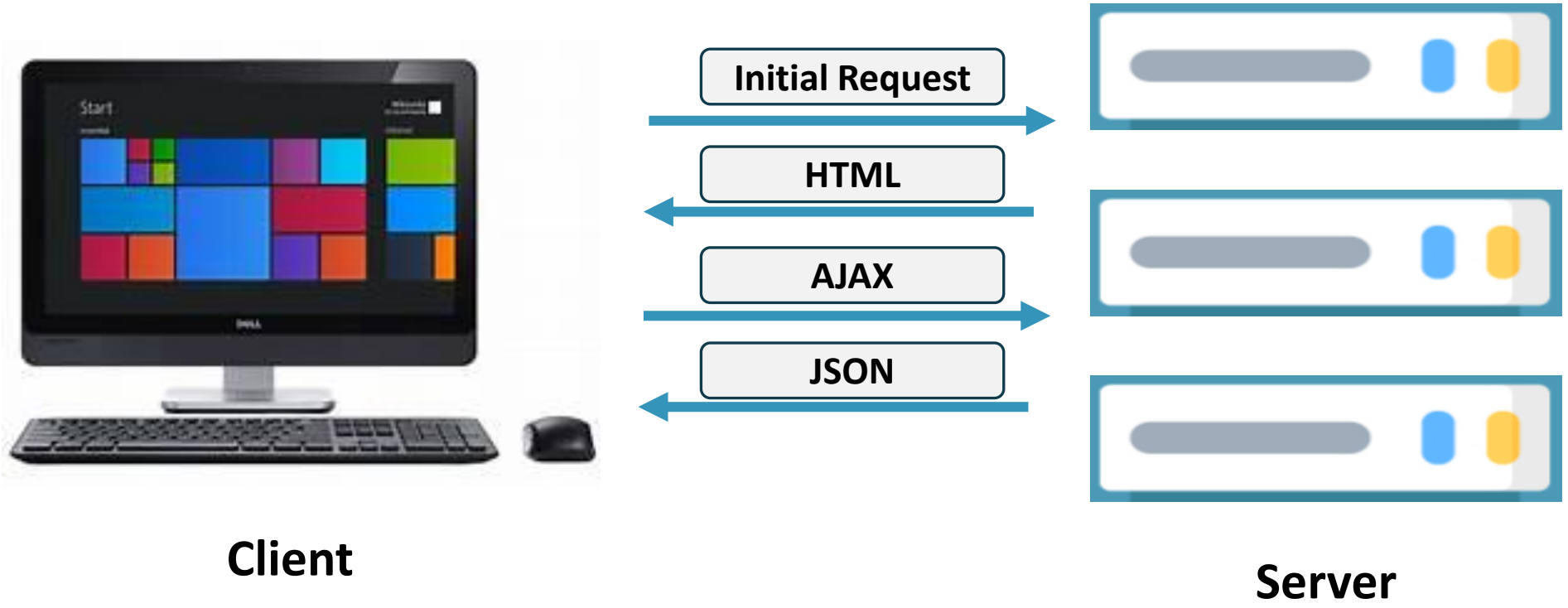
SEO optimization is Tough

- SPAs architecture involves just a single page with a single URL so it loads data once on a server.
- it is hard for the search engines to index the SPA website as opposed to traditional server-rendered pages.

Security Issue

- SPAs applications are less protected from Cross-site scripting (XSS) so hackers can easily insert client side script into web apps.

How do SPAs work ?



Use Cases of SPAs

- E-Commerce Applications
- Social Media Platforms
- Dashboards and Analytics Tools
- Learning Management Systems (LMS)
- Financial and Banking Applications
- Healthcare and Monitoring Systems
- Internal Business Tools

Web Frameworks for developing SPAs

- Ember.js
- Angular.js
- Backbone.js
- Vue.js
- React

Full-stack Development

- What is full stack development ?
- Advantages of Full Stack Web development
- Introduction MEAN Stack
- Components of MEAN stack.
- Setting Up the MEAN Stack.

What is full stack development ?

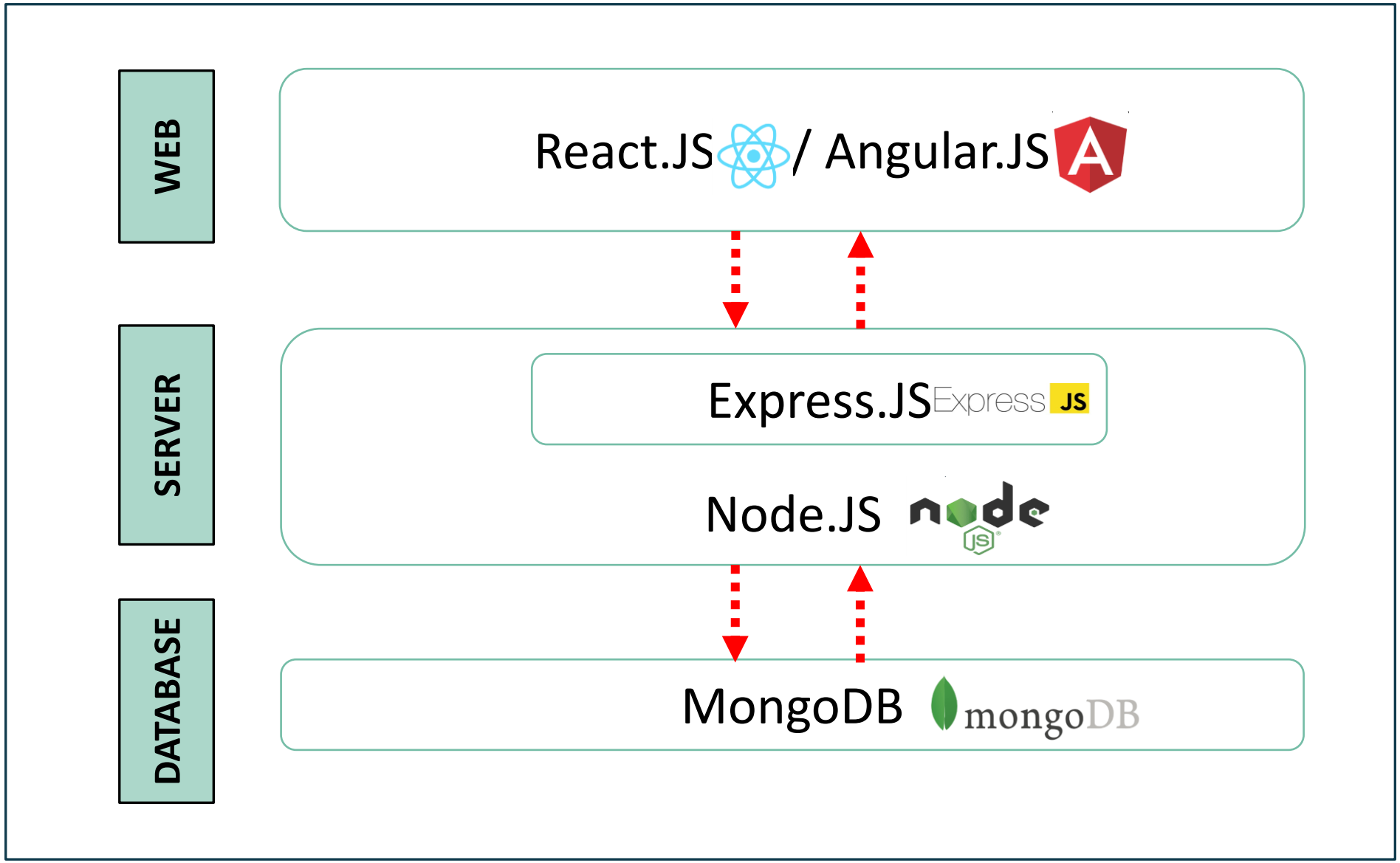
- Full-stack development is the process of planning and creating both the front-end (client-side) and back-end (server-side) elements of a web application.
- For back-end development deals with managing the server-related elements of the website, such as database administration, API development, and more.
- For front-end development concentrates on creating engaging user interfaces and providing a rich user experience.

Advantages of full stack development

- Versality and Flexibility
- Fast development process
- Reduce costs
- Seamless Collaboration
- Complete ownership
- Well wounded solution
- Easy updates

Introduction MEA(R)N Stack

- MEAN stack is open source fully JS-driven, and is widely popular for creating both web applications and cloud-centric solutions.
- It makes it easy to build complex, full-stack web applications and is an excellent option for front and backend development.
- It is a collective technologies that play an important role in the development process of end-to-end web applications.
- It enables to designed to make the development process easier and smoother.


WEB
React.JS / Angular.JS
SERVER
Express.JS
Node.JS
DATABASE
MongoDB

Components of MEAN Stack

MongoDB

- It is the most popular NoSQL (**Non-Structured Query Language**) and open-source document-oriented database.
- The term “NoSQL” refers to a non-relational database that operates without a fixed schema.
- It stores documents in collections using BSON (**Binary JSON**) format.
- It offers a highly scalable, flexible document structure and employs indexing techniques to enhance data retrieval efficiency.
- It is a NoSQL database that scales horizontally by adding more servers and boosts productivity through its flexible document model.

Components of MEAN Stack

ExpressJS

- It is a flexible and powerful JavaScript-based framework for back-end web application development.
- It is primarily used to build single-page applications but can also support multi-page and hybrid web applications.
- It provides robust features for URL routing, middleware integration, and efficient handling of HTTP requests and responses.
- It also makes it easier and simpler to create robust APIs and web servers.
- It is a key component of the MERN / MEAN stacks, used to develop fast, maintainable, and robust production-grade web applications.

Components of MEAN Stack

AngularJS

- Angular is a JavaScript framework used for developing dynamic and interactive front-end applications.
- It is a TypeScript-based web development framework created and maintained by Google.
- It is used to develop scalable single-page and enterprise-level applications tailored to business requirements.
- It offers features such as two-way data binding, dependency injection, testing, and the MVC architecture to create dynamic views and build complex, interactive user interfaces.
- It serves as an extension of the DOM model, enhancing the application's usability and robustness.

Components of MEAN Stack

NodeJS

- It is an open-source, cross-platform runtime environment for executing JavaScript code.
- It allows JavaScript to run on the server side with an event-driven, non-blocking I/O architecture. It is neither a programming language nor a framework.
- Its asynchronous nature allows it to manage multiple concurrent requests efficiently without interrupting the execution of other code.
- It is commonly used to build and develop various backend services, including web and mobile applications.
- It runs on multiple operating systems, including Windows, macOS, Linux, and Unix.

Setting Up the MEAN Stack (MongoDB, Express, Angular, Node.js)

- **Install Node.js and NPM:** Runs JavaScript on the server and installs packages.

  Steps:

  1. Go to https://nodejs.org
  2. Download the LTS version for your operating system.
  3. Install it by following on-screen instructions.
  4. Check installation:

     node –v

     npm -v

Setting Up the MEAN Stack (MongoDB, Express, Angular, Node.js)

- **Install MongoDB:** Database used to store application data.

Steps:

1. Go to https://www.mongodb.com/try/download/community
2. Download and install **MongoDB Community Edition**.
3. Start MongoDB service (it runs automatically).
4. (Optional) Install **MongoDB Compass** — a GUI tool to view your data easily.

Setting Up the MEAN Stack (MongoDB, Express, Angular, Node.js)

- **Set Up Express.js (Backend Framework) :** Helps build web servers and APIs easily.

  npm install express

- **Set Up Angular (Frontend Framework):** Used for creating the user interface.

  npm install -g @angular/cli

Q&
A